# Promila Ghosh

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## EXPERIENCE

**Golden Harvest Infotech** Dhaka, Bangladesh
Senior AI Engineer Sept 2025 – Present

- Led the end-to-end architecture, development, and evaluation of AI-powered RCM development **Hppy Claim** and evaluation for the **USA, UAE, and India markets**, covering **75,000 claims and 82 model runs** with **0% cross-tenant leakage and 0% invalid strategy references**.
- Designed an optimized Denial Intelligence and Strategy Engine by calculating denial patterns in **SQL first**, reducing LLM calls from **15 to 1 per analysis** with **72-hour tenant-aware caching** and deterministic fallback, while sending only aggregated data to the LLM.
- Architected a multimodal healthcare document **HealthScan** intelligence platform, achieving **90% extraction accuracy, 90% agent task success, and 85% retrieval Hit@1** across mixed-format medical documents.
- Drove HealthScan AI to production readiness with **2% OCR CER and 40s p95 latency**, passing the **GO/NO-GO gate with zero safety leaks**.
- Designed a **graph-based patient health knowledge layer** and widget AI agent for continuous patient-context conversations, with a **deterministic fallback layer achieving 100% fallback success**.

**United We Care** July. 2022 – March, 2025
Data Scientist Gurgaon, Haryana, India (Remote)

- Designed and developed a transformer ASR system **United-MedASR** for medical transcription, reducing **Word Error Rate (WER)** to **5.96%** and improving transcription speed by **7-10x** with an optimized interfacing mechanism.
- Built a **5,486 hours** of synthetic clinical audio using a **Diffusion Model** and **GPT**, featuring male and female voices. Open-sourced as **United-Syn-Med** on **Hugging Face** for research.
- Developed a **multi-modal agent system** for **production** using **LangGraph** for analysing and classifying complex medical documents in various formats (*PDFs, scanned images, and handwritten notes*) using a **Large Multi-Modal Model (LMM)**, achieving **70% accuracy** on multi-page and handwritten inputs.
- Built systems for **face detection**, **gesture recognition**, and **facial analysis**, achieving around **60% accuracy** and **faster detection** with **sliding windows** and **Redis** for **real-time processing**
- Streamed real-time user data into **BigQuery** for a **10K+ download app**, improving analytics and data management.

## SKILLS

**Deep Learning** : Neural Networks, Transformers, Large Language Models, Reasoning Models, Vision-Language Models
**LLM & ASR Technologies** : LLaMA, Gemini, GPT, Whisper, Gemma, Retrieval-Augmented Generation (RAG), AI Agents
**Tools & Frameworks** : Python, Scikit-Learn, PyTorch, LangChain, LangGraph, FastAPI, Kafka, Groq, Fireworks AI, Nebius
**Databases & Data** : PostgreSQL, MongoDB, BigQuery, Redis, Amazon S3, Qdrant, Neo4j
**Cloud Computing** : AWS (EC2, Cognito), GCP (Compute Engine, Vertex AI, Google Cloud Storage)

## PROJECTS

- **Medical Document Processing & ASR United We Care** Developed an intelligent system for ASR and medical document analysis using transformers.
- **MediBeng Whisper Tiny: Code-Switched Bengali-English Speech Translation for Clinical Settings** Built a dataset and fine-tuned the Whisper Tiny model for transcription and translation of code-switched speech in healthcare.

## EDUCATION

**M.Sc. Engg., Computer Science and Engineering** February 2022 – August 2023
Khulna University – CGPA: 3.88/4.00 **(Distinction)** Khulna, BD

## Publications

- **Recognition of Sunflower Diseases Using Hybrid Deep Learning and Its Explainability with AI** – *Mathematics (MDPI), 2023*
- **Risk prediction of ischemic heart disease using artificial neural network** – *IEEE, 2018*
- **Multi-labelled Bengali Public Comments Sentiment Analysis with Bidirectional Recurrent Neural Networks** – *Taylor & Francis, 2023*